Stat 517
Homework # 7

1. Let Y_1 and Y_2 be two independent unbiased estimators of θ. Assume that the variance of Y_1 is twice the variance of Y_2. Find the constant k_1 and k_2 so that $k_1 Y_1 + k_2 Y_2$ is an unbiased estimator with the smallest possible variance for such a linear combination.

2. Let $X_1, \ldots, X_n$ be a random sample from a Poisson distribution with parameter θ, $0 < \theta < \infty$. Let $Y = \sum_1^n X_i$ and let $\mathcal{L}\left[\theta, \delta\left(y\right)\right] = \left[\theta - \delta\left(y\right)\right]^2$. If we restrict our considerations to decision functions of the form $\delta\left(y\right) = b + y/n$, where b does not depend on y, show that $R\left(\theta, \delta\right) = b^2 + \theta/n$. What decision function of this form yields a uniformly smaller risk than every other decision function of this form? With this solution, say δ, and $0 < \theta < \infty$, determine $\max_\theta R\left(\theta, \delta\right)$ if it exists.

3. Let $X_1, \ldots, X_n$ be a random sample of size n from a beta distribution with parameters $\alpha = \theta$ and $\beta = 5$. Show that the product $X_1 X_2 \cdots X_n$ is a sufficient statistic for θ.

4. What is the sufficient statistic for θ if the sample arises from a beta distribution in which $\alpha = \beta = \theta > 0$.

5. Let $X_1, \ldots, X_n$ denote a random sample of size $n > 1$ from a distribution with pdf $f(x;\theta) = \theta e^{-\theta x}$, $0 < x < \infty$, zero elsewhere, and $\theta > 0$. Then $Y = \sum_{i=1}^n X_i$ is a sufficient statistic of θ. Prove that $(n-1)/Y$ is the MVUE of θ.

6. Let $X_1, \ldots, X_n$ denote a random sample of size n from a distribution with pdf $f(x;\theta) = \theta x^{\theta - 1}$, $0 < x < 1$, zero elsewhere, and $\theta > 0$.
   (a) Show that the geometric mean $(X_1 X_2 \cdots X_n)^{1/n}$ of the saple is a complete and sufficient statistic of θ.
   (b) Find the MLE of θ, and observe that it is a function of this geometric mean.

7. Let $X_1, \ldots, X_n$ denote a random sample of size n from a distribution with pdf $f(x;\theta) = \theta^2 x e^{-\theta x}$, $0 < x < \infty$, zero elsewhere, and $\theta > 0$.
   (a) Argue that $Y = \sum_{i=1}^n X_i$ is a complete and sufficient statistic for θ.
   (b) Compute $E(1/Y)$ and find the function of Y that is the unique MVUE of θ.

8. Show that the first order statistic Y_1 of a random sample of size n from the distribution having pdf $f(x;\theta) = e^{-(x-\theta)}$, $\theta < x < \infty$, $-\infty < \theta < \infty$, zero elsewhere, is a complete sufficient statistic for θ. Find the unique function of this statistic, which is the MVUE of θ.